



Python Programming Fundamentals

Dictionary Basics

Programming Fundamentals Team

SETU

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Outline

1. Creating a Dictionary
2. Accessing Values
3. Safer Access with `get()`
4. Adding and Updating
5. Useful Dictionary Methods



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1. Creating a Dictionary

```
movie = {  
    "title": "Jaws",  
    "year": 1975,  
    "rating": 5  
}
```



1. Creating a Dictionary

```
movie = {  
    "title": "Jaws",  
    "year": 1975,  
    "rating": 5  
}
```

A dictionary uses:

key: value



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2. Accessing Values

```
print(movie["title"])  
print(movie["year"])
```



2. Accessing Values

```
print(movie["title"])  
print(movie["year"])
```

If the key does not exist, Python raises a `KeyError`.

```
print(movie["director"])
```




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3. Safer Access with get()

```
movie = {"title": "Jaws", "year": 1975}

print(movie.get("title"))
print(movie.get("director"))
print(movie.get("director", "Unknown"))
```



3. Safer Access with get()

```
movie = {"title": "Jaws", "year": 1975}

print(movie.get("title"))
print(movie.get("director"))
print(movie.get("director", "Unknown"))
```

get() is useful when a key might not exist.



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4. Adding and Updating

```
movie = {"title": "Jaws", "year": 1975}

movie["rating"] = 5
movie["year"] = 1976

print(movie)
```



4. Adding and Updating

```
movie = {"title": "Jaws", "year": 1975}

movie["rating"] = 5
movie["year"] = 1976

print(movie)
```

The same syntax is used to add or update.



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5. Useful Dictionary Methods

```
student = {  
    "name": "Ava",  
    "age": 19,  
    "course": "Computing"  
}
```

```
print(student.keys())  
print(student.values())  
print(student.items())
```




5. Useful Dictionary Methods

```
student = {  
    "name": "Ava",  
    "age": 19,  
    "course": "Computing"  
}
```

```
print(student.keys())  
print(student.values())  
print(student.items())
```

Looping over key-value pairs:

```
for key, value in student.items():  
    print(key, value)
```



Thanks for Watching - Any questions?

